

COCOMO-II

It is improved version of original COCOMO and is designed for modern software development including iterative development, reuse and evolving architectures. ①

Why COCOMO-II introduced?

Modern s/w projects introduced several changes:-

- * object oriented development
- * Reusable components
- * GUI builders
- * framework & libraries ... etc

therefore, COCOMO-II was designed to provide more suitable estimation for modern projects.

Types (submodels)

i) Application Composition model

this model is mainly used during early prototyping & rapid application development.

Projects which are having reusable components, in that case it is difficult to count 1000 lines of code. Therefore COCOMO II uses object points [screen, reports, 3GL components].

ii) Early design model

used when project has started to take shape but complete architecture is not yet available. It uses 7 effort multiplier.

iii) Post architecture model

used when the s/w architecture has been developed and enough information is available about project. It uses 17 effort multiplier.

Effort $PM = A \times \text{size}^F \times \prod EM_i$
 $\uparrow$
 2.94

PM = person-month

$A = K = 2.94$

size = s/w size

E = scale exponent

EM = effort multiplier

$\prod EM_i$ = product of all EM

E $\rightarrow$ E represent effort of project scale.

$$E = B + 0.01 \sum SF$$

$$E = 0.91 + 0.01 \sum SF$$

Scale Factor (SF)

it represent characteristics of overall projects that affects how effort grows as s/w become larger.

COCOMO II uses 5 SF.

<u>SF</u>	<u>Full Form</u>	<u>Meaning</u>
PREC	Boundedness	have we developed similar system before.
FLEX	development flexibility	how flexible are requirements
RESL	architecture/risk evolution	how well are risk & architecture understood.
TEAM	team cohesion	how well does team work together
PMAT	Process Maturity	how mature is organization development process

$$\Sigma SF = PREC + FLEX + RESL + TEAM + PHAT$$

then they are used to calculate exponent (E).

$$E = 0.91 + 0.01 \Sigma SF$$

Effort multiplier (EM)

it represents specific condition of product, platform, people or project that can increase or decrease required effort.

The EM are multiplied together as :-

$$\pi EM_i = EM_1 \times EM_2 \times \dots \times EM_n$$

they used in

$$PM = 2.94 \times Size^E \times \pi EM_i$$

EM are grouped in 4 categories as :-

① Product factor

describes characteristics of s/w product. Includes :-

- RELY :- required s/w reliability
- DATA :- DB size
- CPLX :- product complexity
- RUSE :- required reusability
- DOCU :- documentation required.

② Platform factor

describes environment in which s/w operates.

- TIME :- execution time constraint
- STOR :- Main storage constraint
- PVOL :- platform volatility.

③ Personnel Factor

these describe the capabilities of people involved.

ex:-

- ACAP → analyst capability
- PCAP → programmer
- PCON → Personnel continuity
- AEXP → Apps. experience
- PLEX → Platform

④ Project Factor

these describes project environment.

ex:-

- TOOL :- use s/w tools
- SITE :- multisite development
- SCD :- required development schedule

Formula

$$PM = 2.94 \times \text{Size}^E \times TEM_i$$

where

$$E = 0.91 + 0.01 \sum SF$$

Development time

$$TDEV = C(PM)^F$$

where

TDEV = development time in month

$$C = 3.67$$

PM = effort in person-month

$$F = \text{schedule exponent} = 0 + 0.2(E - 0.91) \\ = 0.28 + 0.2(E - 0.91)$$

$$TDEV = 3.67(PM)^F$$

$$\text{Average staff} = \frac{\text{Effort}}{\text{Development time}}$$

$$\text{Q. S/W size} = 50 \text{ KLOC}$$

$$\Sigma SF = 15$$

$$EM = 1$$

Find effort, development time, avg staff

$$E = 0.91 + 0.01 \Sigma SF$$

$$E = 0.91 + 0.01 \times (15)$$

$$= 0.91 + 0.15$$

$$= 1.06$$

$$PM = 2.94 \times \text{Size}^E \times EM \quad [\text{Since } EM = 1]$$

$$= 2.94 \times 50^{1.06}$$

$$PM \approx 174.6 \text{ person/month}$$

Development time

$$F = 0.28 + 0.2 (E - 0.91)$$

$$= 0.28 + 0.2 (1.06 - 0.91)$$

$$= 0.28 + 0.03 = 0.31$$

$$TDEV = 3.67 (PM)^F$$

$$= 3.67 \times (174.6)^{0.31}$$

$$TDEV \approx 18.8 \text{ months}$$

$$\text{Average staff} = \frac{174.6}{18.8}$$

≈ 9.3 person